

Omar Muhammad

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EDUCATION

B-Tech in Mathematics and Computing | 2023 – 2027 | Indian Institute of Science, Bengaluru | CGPA: 9.5/10.0

Grade 12 (CBSE) | 2022 – 2023 | Saraswathi Vidyanikethan, Ernakulam | 98.4% (School Top)

Grade 10 (CBSE) | 2020 – 2021 | Rajagiri Public School, Ernakulam | 98%

PUBLICATIONS

Verification Modulo Tested Library Contracts

Abhishek Uppar, Omar Muhammad, Sumanth Prabhu S, Deepak D'Souza, P. Madhusudan, Adithya Murali.

Accepted to ACM SIGPLAN Conference on Programming Language Design and Implementation (PLDI 2026), to appear in PACMPL. [\[Paper\]](#) [\[Code\]](#)

Mask2Cause: Causal Discovery via Adjacency Constrained Causal Attention

Omar Muhammad, Pasupuleti Dhruv Shrivastava, Deepak N. Subramani

Under review at the conference on Neural Information Processing Systems (NeurIPS 2026). [\[Paper\]](#) [\[Code\]](#)

RESEARCH & INTERNSHIP EXPERIENCE

Summer Research Fellow, Grounded Deduction | EPFL, Switzerland (Onsite) | May 2026 – Present

Adviser: Prof. Bryan Ford, DEDIS Lab

- Formally reasoning about and developing the novel logical framework “Grounded Deduction” in the Isabelle proof assistant.

Student Researcher, Game Theory and Automata | Jan 2026 – Present | IISc

- Investigating supergames played by finite automata under finite costs for complexity to analyze the sequential collapse of equilibrium states.
- Deriving critical phase change thresholds that characterize these equilibrium dynamics.

Student Researcher, Causal Discovery in Time Series | IISc | Aug 2025 – May 2026

Adviser: Prof. Deepak N. Subramani, Computational and Data Sciences, IISc

- Designed a novel Transformer architecture for causal discovery in time series data.
- Developed a trainable attention mask mechanism based on forecasting objectives to learn the underlying causal graph structure, inspired by principles of causal inference.

Student Researcher, Verification Modulo Tested Library Contracts | IISc | May 2025 – Apr 2026

Advisers: Prof. D. D'Souza (IISc), Prof. M. Parthasarathy (UIUC), Prof. A. Murali (UW-Madison)

- Co-developed VMTLC, a hybrid framework that verifies client programs by synthesizing library specifications that are formally proven against the client but validated via testing against the library.
- Designed the Learner component within a Teacher-Learner CEGIS loop, developing two distinct LLM-integrated architectures (a pure LLM learner and an LLM-seeded ICE learner) to leverage semantic intuition, synthesizing valid contracts where traditional enumerative baselines could not.
- Designed and built the core LLM pipeline responsible for generating contracts from logical rules and integrated it into a feedback loop with formal verifiers and testing engines to ensure correctness.

Research Intern, Enhancing Formal Verification with LLMS | University of Virginia, USA (Remote) | Aug 2025 – Oct 2025

Adviser: Prof. Wenzhi Wang, The University of Virginia

- Developed ACPRO, a framework that uses a multi-agent LLM pipeline to transform low-level Z3 proofs into interpretable, high-level Verus verification steps.
- This work also produced the VSVerus dataset, a new resource for studying fine-grained LLM reasoning in formal verification.

Summer Research Fellow, Formalization of Group Theoretic Algorithms | Krea University, India (Onsite) | May 2025 – July 2025

Advisers: Prof. Rishi Vyas (KREA), Prof. T.V.H Prathamesh (KREA)

- Formalized proofs for decision problems in geometric group theory (including the word, conjugacy, and subgroup membership problems) using the Lean 4 proof assistant.
- Key contributions include formalizing two definitions of the Dehn function and proving their equivalence and verifying the algorithm for the conjugacy problem in free groups.
- Current work focuses on the computability of the Dehn function for hyperbolic groups.

Intern | National Payments Corporation of India (NPCI) | Dec 2024 – June 2025

Adviser: Prof. Viraj Kumar, Divecha Centre for Climate Change, IISc

- Developed an internal tool to enable NPCI employees to query databases and generate data visualizations via natural language prompts.
- Implemented a pipeline that translates natural language queries to SQL queries, fetches data, determines correct visualization types, and renders visual output.

Intern | Applied Research Capability Network (ARC-Net), IISc Campus | July 2024 – Jan 2025

- Fine-tuned LLMs with Tool Integrated reasoning for solving math and physics problems.
- Used proof assistants and symbolic solvers to verify the chain of thought of the models.

PROJECTS

Reading Project, Threshold for Random k-SAT | Jan 2026 – April 2026

- Explored the satisfiability threshold for random k-SAT as part of the “Techniques in Discrete Probability” course under the guidance of Prof. Riddhipratim Basu, IISc.
- Analyzed the foundational work by Dimitris Achlioptas and Yuval Peres, on the application of weighted moment methods to bound the threshold. [\[Report\]](#) [\[Presentation\]](#)

Reading Project, PCP Theorem | Aug 2025 – Nov 2025

- Conducted an in-depth study of the PCP theorem, analyzing its formulations in terms of both probabilistically checkable verifiers and the hardness of approximation under the guidance of Prof. Anand Louis, Department of Computer Science and Automation, IISc .
- Analyzed Irit Dinur’s proof of the theorem using gap amplification. [\[Report\]](#)

Reading Project, Undecidability of First-Order Logic | Feb 2025 – April 2025

- Conducted an reading project exploring the undecidability of first-order logic.
- Studied the foundational proof by Alonzo Church, and its implications for computability and formal verification systems.

Research Project, Image Denoising Algorithms | March 2025 – April 2025

- Implemented and evaluated various frameworks for image restoration without clean data.
- Designed and conducted experiments to analyze the hyperparameter sensitivity of Zero-Shot Noise2Noise implementation. [\[Report\]](#) [\[Presentation\]](#)

Formalization Project, Rubik’s Cube Solvability Conditions | Feb 2025 – May 2025

- Modeled the nxn Rubik’s Cube as a group, formalized proof of validity of solvability conditions of the $4 \times 4 \times 4$ Rubik’s Cube and formulated the solvability conditions for the nxn cube in Lean 4. [\[Github Repo\]](#)

Robotic Arm Motion Planning Lead | Dec 2024 – March 2025

- Responsible for motion planning for the robotic arm of the rover using MoveIt2 and ROS 2.

Reading project, Differential Topology and Knot Theory | May 2024 – July 2024

- Explored topics in Differential Topology and Knot Theory under the guidance of Prof. Subhojoy Gupta, Department of Mathematics, Indian Institute of Science.

Developer DataBased (CS club at IISc) | Oct 2023 – Dec 2023

- Utilized Generative AI to develop components for an automated Refute Problem Generation system.

ACHIEVEMENTS AND RECOGNITIONS

- Selected for MITACS Globalink Research Internship 2026 (Offers: Undergraduate Summer Research Program [UGSRP] at University of Toronto; Fields Undergraduate Summer Research Program [FUSRP] at University of Waterloo)
- JEE Advanced Examination: 646th rank
- JEE Mains Examination: 99.95 percentile, 654th rank
- KVPY (Kishore Vaigyanik Protsahan Yojana) Scholar
- NTSE (National Talent Search Examination) Scholar
- NSEP (National Standard Examination in Physics, IAPT) Qualified, National Top 1%
- NSEA (National Standard Examination in Astronomy, IAPT) Qualified
- NSEC (National Standard Examination in Chemistry, ACT) Qualified
- IOQM (Indian Olympiad Qualifier in Mathematics) Qualified

TECHNICAL SKILLS

- **Verification & Logic:** Lean 4, Z3, SMT Solvers, Alloy, Spin, Rodin, VCC, JPF, Rodin, Verus, Formal Methods
- **Mathematics:** Statistics, Probability, Topology, Abstract Algebra, Real Analysis, Linear Algebra, Combinatorics, Graph Theory, Number Theory, Multivariable Calculus, Stochastic Processes
- **Theoretical Computer Science:** Automata Theory, Game Theory & Mechanism Design, Program Analysis & Verification, Numerical Methods, Probabilistic Methods, Information Theory Discrete Fourier Analysis, Convex Optimization
- **Machine Learning:** PyTorch, Scikit-learn, Transformers, LLM Fine-Tuning, Causal Discovery
- **Languages & Tools:** Python, C, SQL

EXTRACURRICULAR ACTIVITIES

- Mentored underprivileged students in STEM concepts through the Notebook Drive Mentorship Outreach Program, IISc.
- Active Member of DataBased (Computer Science Club at IISc).
- Designed competitive programming problems for the RevCoding Contest as a Volunteer for Pravega 2025 (IISc UG Fest).
- Presented on 'Online Decision-Making Algorithms' at Algorithm Festival 2024, IISc.
- Enthusiastic participant in MUN (Model United Nations) conferences and debate competitions.

LANGUAGES

- **Fluent:** English, Malayalam
- **Intermediate:** Hindi